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import random
import time

def print_grid(grid):
    """Displays the board. 0-8 layout."""
    print(f"\n {grid[0]} | {grid[1]} | {grid[2]} ")
    print("-----")
    print(f" {grid[3]} | {grid[4]} | {grid[5]} ")
    print("-----")
    print(f" {grid[6]} | {grid[7]} | {grid[8]} \n")

def check_win(grid, player):
    """Checks rows, columns, and diagonals."""
    lines = [
        (0,1,2), (3,4,5), (6,7,8), # Rows
        (0,3,6), (1,4,7), (2,5,8), # Cols
        (0,4,8), (2,4,6)           # Diagonals
    ]
    for a,b,c in lines:
        if grid[a] == grid[b] == grid[c] == player:
            return True
    return False

def get_ai_move(grid):
    """AI Logic: 1. Win? 2. Block Human? 3. Random."""
    possible_moves = [i for i, spot in enumerate(grid) if spot == ' ']

    # 1. Try to WIN immediately
    for move in possible_moves:
        grid_copy = grid[:]
        grid_copy[move] = 'O'
        if check_win(grid_copy, 'O'):
            return move

    # 2. Block YOU if you are about to win
    for move in possible_moves:
        grid_copy = grid[:]
        grid_copy[move] = 'X'
        if check_win(grid_copy, 'X'):
            return move

    # 3. Pick Random
    return random.choice(possible_moves)

def play_game():
    grid = [' ']*9
    print("Name: HRITHIKA RAJESH")
    print("Usn: 1BM24CS120")
    print("Tic-Tac-Toe: You are X, AI is O.")
    print("Enter positions 0-8 to play.")

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print("Enter positions 0-8 to play.")
print_grid(grid)

while True:
    # --- HUMAN TURN (X) ---
    while True:
        try:
            user_input = int(input("Your Move (X): "))
            if 0 <= user_input <= 8 and grid[user_input] == ' ':
                grid[user_input] = 'X'
                break # Valid move, exit input loop
            else:
                print("Position taken or invalid. Try again.")
        except ValueError:
            print("Please enter a number 0-8.")

    print_grid(grid)

    if check_win(grid, 'X'):
        print("🎉 YOU WIN! 🎉")
        break
    if ' ' not in grid:
        print("Draw Game!")
        break

    # --- AI TURN (O) ---
    # No input() here. It runs automatically.
    print("AI (O) is thinking...")
    time.sleep(1) # Small delay so it feels like thinking

    ai_move = get_ai_move(grid)
    grid[ai_move] = 'O' # AI places move

    print(f"AI plays at position: {ai_move}")
    print_grid(grid)

    if check_win(grid, 'O'):
        print("🤖 AI WINS! 🤖")
        break
    if ' ' not in grid:
        print("Draw Game!")
        break

if __name__ == "__main__":
    play_game()

```

Name: HRITHIKA RAJESH

Usn: 1BM24CS120

Tic-Tac-Toe: You are X, AI is O.

Enter positions 0-8 to play.

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|   |
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|   |

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Your Move (X): 0

```

X |   |
-----
|   |
-----
|   |

```

AI (O) is thinking...
AI plays at position: 5

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X |   |
-----
|   | O
-----
|   |

```

Your Move (X): 4

```

X |   |
-----
| X | O
-----
|   |

```

AI (O) is thinking...
AI plays at position: 8

```

X |   |
-----
| X | O
-----
|   | O

```

Your Move (X): 2

```

X |   | X
-----
| X | O
-----
|   | O

```

AI (O) is thinking...
AI plays at position: 1

```

X | O | X

```

